

Problem-Solving & Data

ANSWER KEY



Ratios and proportions

- 1 A recipe calls for 3 cups of flour for every 2 cups of sugar. If a baker uses 9 cups of flour, how many cups of sugar are needed?

Set up a proportion: $\frac{3}{2} = \frac{9}{x}$. Cross-multiply: $3x = 18$, so $x = 6$ cups of sugar.

- 2 On a map, 1 inch represents 25 miles. If two cities are 7 inches apart on the map, what is the actual distance between them?

$$7 \times 25 = 175 \text{ miles.}$$

Percent problems

- 3 A shirt originally priced at 40 dollars is on sale for 25 percent off. Find the sale price.

Discount amount: $40 \times 0.25 = 10$ dollars. Sale price: $40 - 10 = 30$ dollars.

- 4 A population increased from 4000 to 4600. Find the percent increase.

Increase: $4600 - 4000 = 600$. Percent increase: $\frac{600}{4000} \times 100\% = 15\%$.

Unit conversions and rates

- 5 A car travels at 60 miles per hour. How many miles does it travel in 45 minutes?

45 minutes = 0.75 hours. Distance = $60 \times 0.75 = 45$ miles.

- 6 A factory produces 480 units in an 8-hour shift. Find the production rate in units per minute.

8 hours = 480 minutes. Rate = $\frac{480 \text{ units}}{480 \text{ minutes}} = 1$ unit per minute.

Reading and interpreting scatterplots and tables

- 7 A table shows study hours and test scores: (2,65), (4,75), (6,85), (8,95). Describe the relationship between study hours and test score, and predict the score for 5 hours of study.

The relationship is linear: each additional 2 hours of study increases the score by 10 points, giving a rate of 5 points per hour. Using the pattern, for 5 hours: starting from (4,75), add half of the 2-hour increment (5 points): $75 + 5 = 80$.

Measures of center: mean, median, and mode

- 8 Find the mean, median, and mode of the data set: 4, 7, 7, 9, 12, 15.

Mean: $\frac{4 + 7 + 7 + 9 + 12 + 15}{6} = \frac{54}{6} = 9$. Median (average of the two middle values, 7 and 9): $\frac{7 + 9}{2} = 8$. Mode: 7 (appears twice, more than any other value).

- 9 A data set has mean 20 and 5 values. If four of the values are 15, 18, 22, 25, find the fifth value.

Sum of all 5 values = $20 \times 5 = 100$. Sum of the known 4 values = $15 + 18 + 22 + 25 = 80$. Fifth value = $100 - 80 = 20$.

Measures of spread and the effect of outliers

- 10 Find the range and explain the effect on the mean if an outlier of 50 is added to the data set 10, 12, 14, 16, 18.

Original range: $18 - 10 = 8$. Original mean: $\frac{10 + 12 + 14 + 16 + 18}{5} = \frac{70}{5} = 14$. Adding the outlier 50: new mean = $\frac{70 + 50}{6} = \frac{120}{6} = 20$. The outlier significantly increases the mean (from 14 to 20) and greatly increases the range (from 8 to 40), showing that the mean and range are both highly sensitive to outliers, unlike the median.

Probability from data tables

- 11 A survey of 200 students found 120 prefer math and 80 prefer science. If a student is selected at random, find the probability the student prefers math.

$$P(\text{math}) = \frac{120}{200} = 0.6, \text{ or } 60\%.$$

- 12 A two-way table shows: 50 students play sports and pass an exam; 20 play sports and fail; 30 don't play sports and pass; 10 don't play sports and fail. Find the probability a randomly selected student plays sports, given the student passed the exam.

Total who passed: $50 + 30 = 80$. Of those, 50 play sports. Conditional probability: $\frac{50}{80} = 0.625$, or 62.5%.

Population, sample, and margin of error

- 13** A poll of 500 randomly selected voters found that 55% support a proposal, with a margin of error of 4%. Interpret this result and state the range of plausible values for the true population percentage.

The margin of error means the true percentage of all voters (the population) who support the proposal is likely between $55\% - 4\% = 51\%$ and $55\% + 4\% = 59\%$, based on this sample of 500 voters. It reflects the uncertainty introduced by surveying only a sample rather than the entire population.

Weighted averages

- 14** A student's final grade is weighted: homework counts 20%, tests count 50%, and the final exam counts 30%. If the student scores 90 on homework, 80 on tests, and 85 on the final exam, find the weighted final grade.

$$\text{Weighted grade} = 0.20(90) + 0.50(80) + 0.30(85) = 18 + 40 + 25.5 = 83.5.$$

- 15** A store mixes 3 pounds of coffee costing 8 dollars per pound with 2 pounds of coffee costing 13 dollars per pound. Find the price per pound of the mixture.

$$\begin{array}{l} \text{Total cost: } 3(8) + 2(13) = 24 + 26 = 50 \text{ dollars. Total weight: } 3 + 2 = 5 \text{ pounds. Price per pound:} \\ \frac{50}{5} = 10 \text{ dollars.} \end{array}$$

Multi-step unit conversion problems

- 16** A machine produces 15 items per minute. Find the number of items produced in 3.5 hours.

$$\text{Convert hours to minutes: } 3.5 \times 60 = 210 \text{ minutes. Total items: } 15 \times 210 = 3150 \text{ items.}$$

- 17** A car uses fuel at a rate of 28 miles per gallon. If gas costs 3.50 dollars per gallon, find the fuel cost for a 420-mile trip.

$$\text{Gallons needed: } \frac{420}{28} = 15 \text{ gallons. Fuel cost: } 15 \times 3.50 = 52.50 \text{ dollars.}$$

Percent change applied twice (successive changes)

- 18** A stock price increases by 20% one year, then decreases by 20% the next year. If the original price was 100 dollars, find the final price, and explain why it is not back to 100 dollars.

After the increase: $100(1.20) = 120$ dollars. After the decrease: $120(0.80) = 96$ dollars. The final price is not back to the original because the percent decrease is applied to a larger base (120) than the percent increase was applied to (100), so the dollar amount lost in the decrease (24) exceeds the dollar amount gained in the increase (20).

Density and rate-based problem solving

- 19** A population density is 150 people per square mile. If a city has an area of 24 square miles, find the total population.

Population = $150 \times 24 = 3600$ people.

- 20** Three workers can complete a job in 12 hours together. If they work at the same combined rate, how many hours would it take 2 workers with the same total capacity to complete a job that is 1.5 times as large?

Combined rate for 3 workers: $\frac{1 \text{ job}}{12 \text{ hours}}$. For 2 workers with proportionally less capacity (assuming equal individual rates), the rate scales down to $\frac{2}{3}$ of the original: $\frac{2}{3} \times \frac{1}{12} = \frac{1}{18}$ jobs per hour. Time for 1.5 jobs: $1.5 \div \frac{1}{18} = 1.5 \times 18 = 27$ hours.

Multiple choice — easy 15

- 21** A recipe calls for 4 cups of flour for every 9 cups of sugar. If a baker uses 32 cups of flour, how many cups of sugar are needed?

- a. 80
- b. 32
- c. 64
- d. 72 ✓

Set up a proportion: $4/9 = 32/x$. Cross-multiply: $4x = 288$, so $x = 72$ cups of sugar.

- 22** A recipe calls for 7 cups of rice for every 2 cups of beans. If a baker uses 77 cups of rice, how many cups of beans are needed?

- a. 77
- b. 33
- c. 11
- d. 22 ✓

Set up a proportion: $7/2 = 77/x$. Cross-multiply: $7x = 154$, so $x = 22$ cups of beans.

- 23** A recipe calls for 5 cups of water for every 2 cups of concentrate. If a baker uses 20 cups of water, how many cups of concentrate are needed?
- a. 4
 - b. 12
 - c. 20
 - d. 8 ✓

Set up a proportion: $\frac{5}{2} = \frac{20}{x}$. Cross-multiply: $5x = 40$, so $x = 8$ cups of concentrate.

- 24** A recipe calls for 5 cups of water for every 4 cups of concentrate. If a baker uses 15 cups of water, how many cups of concentrate are needed?
- a. 8
 - b. 15
 - c. 12 ✓
 - d. 9

Set up a proportion: $\frac{5}{4} = \frac{15}{x}$. Cross-multiply: $5x = 60$, so $x = 12$ cups of concentrate.

- 25** A recipe calls for 9 cups of red paint for every 3 cups of blue paint. If a baker uses 72 cups of red paint, how many cups of blue paint are needed?
- a. 32
 - b. 16
 - c. 24 ✓
 - d. 72

Set up a proportion: $\frac{9}{3} = \frac{72}{x}$. Cross-multiply: $9x = 216$, so $x = 24$ cups of blue paint.

- 26 A recipe calls for 8 cups of red paint for every 3 cups of blue paint. If a baker uses 48 cups of red paint, how many cups of blue paint are needed?
- a. 12
 - b. 48
 - c. 18 ✓
 - d. 24

Set up a proportion: $8/3 = 48/x$. Cross-multiply: $8x = 144$, so $x = 18$ cups of blue paint.

- 27 A recipe calls for 5 cups of rice for every 9 cups of beans. If a baker uses 35 cups of rice, how many cups of beans are needed?
- a. 63 ✓
 - b. 56
 - c. 35
 - d. 70

Set up a proportion: $5/9 = 35/x$. Cross-multiply: $5x = 315$, so $x = 63$ cups of beans.

- 28 A recipe calls for 8 cups of cement for every 5 cups of sand. If a baker uses 96 cups of cement, how many cups of sand are needed?
- a. 60 ✓
 - b. 96
 - c. 72
 - d. 48

Set up a proportion: $8/5 = 96/x$. Cross-multiply: $8x = 480$, so $x = 60$ cups of sand.

- 29** A recipe calls for 6 cups of flour for every 2 cups of sugar. If a baker uses 36 cups of flour, how many cups of sugar are needed?
- a. 18
 - b. 6
 - c. 12 ✓
 - d. 36

Set up a proportion: $\frac{6}{2} = \frac{36}{x}$. Cross-multiply: $6x = 72$, so $x = 12$ cups of sugar.

- 30** A recipe calls for 7 cups of red paint for every 4 cups of blue paint. If a baker uses 28 cups of red paint, how many cups of blue paint are needed?
- a. 28
 - b. 16 ✓
 - c. 12
 - d. 20

Set up a proportion: $\frac{7}{4} = \frac{28}{x}$. Cross-multiply: $7x = 112$, so $x = 16$ cups of blue paint.

- 31** A recipe calls for 7 cups of flour for every 5 cups of sugar. If a baker uses 56 cups of flour, how many cups of sugar are needed?
- a. 32
 - b. 56
 - c. 48
 - d. 40 ✓

Set up a proportion: $\frac{7}{5} = \frac{56}{x}$. Cross-multiply: $7x = 280$, so $x = 40$ cups of sugar.

- 32** A recipe calls for 8 cups of rice for every 5 cups of beans. If a baker uses 40 cups of rice, how many cups of beans are needed?

a. 25 ✓

b. 40

c. 20

d. 30

Set up a proportion: $8/5 = 40/x$. Cross-multiply: $8x = 200$, so $x = 25$ cups of beans.

- 33** A recipe calls for 9 cups of rice for every 5 cups of beans. If a baker uses 27 cups of rice, how many cups of beans are needed?

a. 15 ✓

b. 12

c. 18

d. 27

Set up a proportion: $9/5 = 27/x$. Cross-multiply: $9x = 135$, so $x = 15$ cups of beans.

- 34** A recipe calls for 8 cups of water for every 4 cups of concentrate. If a baker uses 32 cups of water, how many cups of concentrate are needed?

a. 32

b. 20

c. 16 ✓

d. 12

Set up a proportion: $8/4 = 32/x$. Cross-multiply: $8x = 128$, so $x = 16$ cups of concentrate.

- 35 A recipe calls for 5 cups of red paint for every 2 cups of blue paint. If a baker uses 40 cups of red paint, how many cups of blue paint are needed?
- a. 40
 - b. 16 ✓
 - c. 24
 - d. 8

Set up a proportion: $\frac{5}{2} = \frac{40}{x}$. Cross-multiply: $5x = 80$, so $x = 16$ cups of blue paint.

Multiple choice — medium 15

- 36 A desk originally priced at 70 dollars is on sale for 40 percent off. Find the sale price.
- a. 42 ✓
 - b. 47
 - c. 28
 - d. 37

Discount amount: $70 \times 0.40 = 28$ dollars. Sale price: $70 - 28 = 42$ dollars.

- 37 A laptop originally priced at 105 dollars is on sale for 20 percent off. Find the sale price.
- a. 84 ✓
 - b. 89
 - c. 21
 - d. 79

Discount amount: $105 \times 0.20 = 21$ dollars. Sale price: $105 - 21 = 84$ dollars.

38 A bicycle originally priced at 180 dollars is on sale for 10 percent off. Find the sale price.

- a. 157
- b. 18
- c. 167
- d. 162 ✓

Discount amount: $180 \times 0.10 = 18$ dollars. Sale price: $180 - 18 = 162$ dollars.

39 A watch originally priced at 200 dollars is on sale for 25 percent off. Find the sale price.

- a. 50
- b. 150 ✓
- c. 155
- d. 145

Discount amount: $200 \times 0.25 = 50$ dollars. Sale price: $200 - 50 = 150$ dollars.

40 A bicycle originally priced at 160 dollars is on sale for 35 percent off. Find the sale price.

- a. 99
- b. 104 ✓
- c. 56
- d. 109

Discount amount: $160 \times 0.35 = 56$ dollars. Sale price: $160 - 56 = 104$ dollars.

41 A laptop originally priced at 20 dollars is on sale for 30 percent off. Find the sale price.

- a. 6
- b. 19
- c. 9
- d. 14 ✓

Discount amount: $20 \times 0.30 = 6$ dollars. Sale price: $20 - 6 = 14$ dollars.

42 A television originally priced at 200 dollars is on sale for 15 percent off. Find the sale price.

- a. 30
- b. 175
- c. 165
- d. 170 ✓

Discount amount: $200 \times 0.15 = 30$ dollars. Sale price: $200 - 30 = 170$ dollars.

43 A laptop originally priced at 185 dollars is on sale for 20 percent off. Find the sale price.

- a. 148 ✓
- b. 37
- c. 143
- d. 153

Discount amount: $185 \times 0.20 = 37$ dollars. Sale price: $185 - 37 = 148$ dollars.

44 A laptop originally priced at 110 dollars is on sale for 20 percent off. Find the sale price.

- a. 88 ✓
- b. 83
- c. 93
- d. 22

Discount amount: $110 \times 0.20 = 22$ dollars. Sale price: $110 - 22 = 88$ dollars.

45 A bicycle originally priced at 75 dollars is on sale for 40 percent off. Find the sale price.

a. 45 ✓

b. 50

c. 30

d. 40

Discount amount: $75 \times 0.40 = 30$ dollars. Sale price: $75 - 30 = 45$ dollars.

46 A backpack originally priced at 120 dollars is on sale for 5 percent off. Find the sale price.

a. 114 ✓

b. 119

c. 109

d. 6

Discount amount: $120 \times 0.05 = 6$ dollars. Sale price: $120 - 6 = 114$ dollars.

47 A desk originally priced at 130 dollars is on sale for 10 percent off. Find the sale price.

a. 13

b. 122

c. 117 ✓

d. 112

Discount amount: $130 \times 0.10 = 13$ dollars. Sale price: $130 - 13 = 117$ dollars.

48 A backpack originally priced at 30 dollars is on sale for 40 percent off. Find the sale price.

a. 12

b. 18 ✓

c. 13

d. 23

Discount amount: $30 \times 0.40 = 12$ dollars. Sale price: $30 - 12 = 18$ dollars.

49 A television originally priced at 80 dollars is on sale for 35 percent off. Find the sale price.

- a. 57
- b. 47
- c. 52 ✓
- d. 28

Discount amount: $80 \times 0.35 = 28$ dollars. Sale price: $80 - 28 = 52$ dollars.

50 A laptop originally priced at 40 dollars is on sale for 15 percent off. Find the sale price.

- a. 6
- b. 29
- c. 34 ✓
- d. 39

Discount amount: $40 \times 0.15 = 6$ dollars. Sale price: $40 - 6 = 34$ dollars.

Multiple choice — hard 15

51 A student's final grade is weighted: homework counts 30%, tests count 35%, and the final exam counts 35%. If the student scores 78 on homework, 65 on tests, and 90 on the final exam, find the weighted final grade.

- a. 77.67
- b. 74.65
- c. 77.65 ✓
- d. 80.65

Weighted grade = $0.30(78) + 0.35(65) + 0.35(90) = 77.65$.

- 52 A student's final grade is weighted: homework counts 15%, tests count 35%, and the final exam counts 50%. If the student scores 80 on homework, 93 on tests, and 73 on the final exam, find the weighted final grade.
- a. 84.05
 - b. 82.00
 - c. 81.05 ✓
 - d. 78.05

$$\text{Weighted grade} = 0.15(80) + 0.35(93) + 0.50(73) = 81.05.$$

- 53 A student's final grade is weighted: homework counts 20%, tests count 50%, and the final exam counts 30%. If the student scores 80 on homework, 98 on tests, and 66 on the final exam, find the weighted final grade.
- a. 81.33
 - b. 84.80 ✓
 - c. 87.80
 - d. 81.80

$$\text{Weighted grade} = 0.20(80) + 0.50(98) + 0.30(66) = 84.80.$$

- 54 A student's final grade is weighted: homework counts 15%, tests count 35%, and the final exam counts 50%. If the student scores 86 on homework, 82 on tests, and 91 on the final exam, find the weighted final grade.
- a. 84.10
 - b. 87.10 ✓
 - c. 90.10
 - d. 86.33

$$\text{Weighted grade} = 0.15(86) + 0.35(82) + 0.50(91) = 87.10.$$

- 55 A student's final grade is weighted: homework counts 10%, tests count 35%, and the final exam counts 55%. If the student scores 72 on homework, 87 on tests, and 94 on the final exam, find the weighted final grade.
- a. 92.35
 - b. 86.35
 - c. 84.33
 - d. 89.35 ✓

$$\text{Weighted grade} = 0.10(72) + 0.35(87) + 0.55(94) = 89.35.$$

- 56 A two-way table shows: 32 students take the bus and arrive on time; 19 take the bus but do not arrive on time; 33 do not take the bus but do arrive on time; 33 do neither. Find the probability a randomly selected student takes the bus, given that the student arrives on time.
- a. 0.49 ✓
 - b. 0.39
 - c. 0.59
 - d. 0.63

Total who arrive on time: $32 + 33 = 65$. Of those, 32 also take the bus. Conditional probability:
 $32/65 \approx 0.49$.

- 57 A two-way table shows: 47 students own a car and commute over 30 minutes; 38 own a car but do not commute over 30 minutes; 39 do not own a car but do commute over 30 minutes; 17 do neither. Find the probability a randomly selected student owns a car, given that the student commutes over 30 minutes.
- a. 0.65
 - b. 0.55 ✓
 - c. 0.45
 - d. 0.60

Total who commute over 30 minutes: $47 + 39 = 86$. Of those, 47 also own a car. Conditional probability:
 $47/86 \approx 0.55$.

- 58 A two-way table shows: 31 students like coffee and like tea; 25 like coffee but do not like tea; 36 do not like coffee but do like tea; 34 do neither. Find the probability a randomly selected student likes coffee, given that the student likes tea.

a. 0.36
b. 0.56
c. 0.55
d. 0.46 ✓

Total who like tea: $31 + 36 = 67$. Of those, 31 also like coffee. Conditional probability: $31/67 \approx 0.46$.

- 59 A two-way table shows: 31 students take the bus and arrive on time; 30 take the bus but do not arrive on time; 13 do not take the bus but do arrive on time; 34 do neither. Find the probability a randomly selected student takes the bus, given that the student arrives on time.

a. 0.70 ✓
b. 0.51
c. 0.80
d. 0.60

Total who arrive on time: $31 + 13 = 44$. Of those, 31 also take the bus. Conditional probability: $31/44 \approx 0.70$.

- 60 A two-way table shows: 26 students like coffee and like tea; 36 like coffee but do not like tea; 5 do not like coffee but do like tea; 36 do neither. Find the probability a randomly selected student likes coffee, given that the student likes tea.

a. 0.42
b. 0.74
c. 0.94
d. 0.84 ✓

Total who like tea: $26 + 5 = 31$. Of those, 26 also like coffee. Conditional probability: $26/31 \approx 0.84$.

61 A price of 200 dollars increases by 25%, then the new price increases by 25%. Find the final price.

- a. 200.00
- b. 312.50 ✓
- c. 317.50
- d. 307.50

After the first change: $200(1.25) = 250.00$ dollars. After the second change: $250.00(1.25) = 312.50$ dollars.

62 A price of 120 dollars increases by 10%, then the new price increases by 10%. Find the final price.

- a. 150.20
- b. 140.20
- c. 120.00
- d. 145.20 ✓

After the first change: $120(1.10) = 132.00$ dollars. After the second change: $132.00(1.10) = 145.20$ dollars.

63 A price of 120 dollars decreases by 30%, then the new price increases by 25%. Find the final price.

- a. 100.00
- b. 105.00 ✓
- c. 110.00
- d. 120.00

After the first change: $120(0.70) = 84.00$ dollars. After the second change: $84.00(1.25) = 105.00$ dollars.

64 A price of 120 dollars increases by 25%, then the new price decreases by 30%. Find the final price.

- a. 110.00
- b. 100.00
- c. 120.00
- d. 105.00 ✓

After the first change: $120(1.25) = 150.00$ dollars. After the second change: $150.00(0.70) = 105.00$ dollars.

65 A price of 120 dollars decreases by 10%, then the new price increases by 15%. Find the final price.

- a. 120.00
- b. 119.20
- c. 124.20 ✓
- d. 129.20

After the first change: $120(0.90) = 108.00$ dollars. After the second change: $108.00(1.15) = 124.20$ dollars.